$$\frac{P^0_{solvent} - P_{solution}}{P^0_{solvent}} = x_B$$

Let w_A and w_B be the weights of the solvent and solute respectively and their corresponding molar masses are M_A and M_B, then the mole fraction of the solute x_B is

$$x_B = \frac{n_B}{n_A + n_B} \qquad (9.20)$$

Here, n_A & n_B are the moles of the solvent and the solute respectively. For dilute solutions $n_A >> n_B$. Hence $n_A + n_B \approx n_A$. Now

$$x_B = \frac{n_B}{n_A}$$

Number of moles of solvent and the solute are,

$$n_A = \frac{w_A}{M_A}, \; n_B = \frac{w_B}{M_B}$$

$$\text{Therefore, } x_B = \frac{\frac{w_B}{M_B}}{\frac{w_A}{M_A}} \quad \text{------ (9.21)}$$

Thus,

$$\frac{P^\circ_{solvent} - P_{solution}}{P^\circ_{solvent}} = \frac{\frac{w_B}{M_B}}{\frac{w_A}{M_A}}$$

$$\frac{\Delta P}{P^\circ_A} = \frac{w_B \times M_A}{w_A \times M_B} \quad \text{------ (9.22)}$$

From the equation (9.22) the molar mass of the solute (M_B) can be calculated using the known values of w_A, w_B, M_A and the measured relative lowering of vapour pressure.

Example Problem3:

An aqueous solution of 2 % nonvolatile solute exerts a pressure of 1.004 bar at the boiling point of the solvent. What is the molar mass of the solute when P°_A is 1.013 bar?

$$\frac{\Delta P}{P^\circ_A} = \frac{W_B \text{ x } M_A}{M_B \text{ x } W_A}$$

In a 2 % solution weight of the solute is 2 g and solvent is 98 g

$\Delta P = P^\circ_A - P_{solution} = 1.013 - 1.004$ bar $= 0.009$ bar

$$M_B = \frac{P^\circ_A \text{ x } W_B \text{ x} M_A}{\Delta P \text{ x } W_A}$$

$M_B = 2 \text{ x } 18 \text{ x } 1.013/(98 \text{ x } 0.009)$

$= 41.3 \text{ g mol}^{-1}$

Evaluate Yourself

10) Vapour pressure of a pure liquid A is 10.0 torr at 27°C . The vapour pressure is lowered to 9.0 torr on dissolving one gram of B in 20 g of A. If the molar mass of A is 200 g mol^{-1} then calculate the molar mass of B.

Elevation of boiling point

Boiling point is an important physical property of a liquid. The boiling point of a liquid is the temperature at which its vapour pressure becomes equal to the atmospheric pressure (1 atm). When a nonvolatile solute is added to a pure solvent at its boiling point, the vapour pressure of the solution is lowered below 1 atm. To bring the vapour pressure again to 1 atm, the temperature of the solution has to be increased. As a result,the solution boils at a higher temperature (T_b) than the boiling point of the pure solvent (T°_b). This increase

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